

- Consider an $N \times N$ random matrix J_{ij}

- with $P(J_{ij}) \sim \begin{cases} \mathcal{N}(0, \sigma^2/N) & i \neq j \\ \mathcal{N}(0, \frac{2\sigma^2}{N}) & i = j \end{cases}$

this normalization of variances allows
unitary invariance

- then $P(J) \propto e^{-\frac{N}{4\sigma^2} \text{Tr } J^T J} \rightarrow P(UJ U^T) = P(J) \Rightarrow$ prob of J
depends only
on eigenvalues
of J

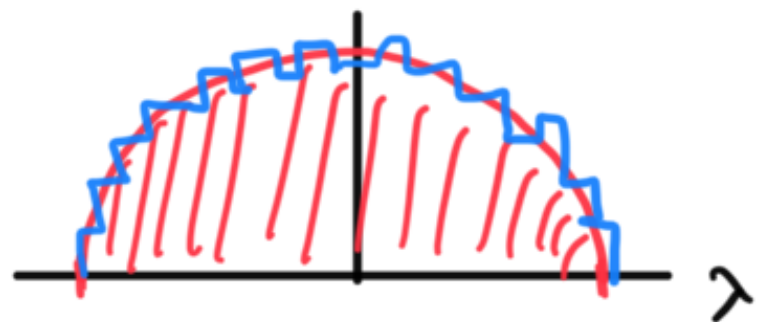
$$\begin{aligned}
 &= e^{-\frac{N}{4\sigma^2} \sum_{i,j} J_{ij}^2} \\
 &= e^{-\frac{N}{2\sigma^2} \sum_{i < j} J_{ij}^2 - \frac{N}{4\sigma^2} \sum_i J_{ii}^2} \\
 &\quad \underbrace{\text{Var}(J_{ij}) = \frac{\sigma^2}{N}}_{i \neq j} \quad \underbrace{\text{Var}(J_{ii}) = \frac{2\sigma^2}{N}}
 \end{aligned}$$

- let $\lambda_1, \dots, \lambda_N$ be the eigenvalues of J

- $\rho_J(\lambda) \equiv \frac{1}{N} \sum_{i=1}^N \delta[\lambda - \lambda_i]$ empirical eigenvalue distribution of J

- for large N $\rho_J(\lambda)$ (ie empirical histogram of eigenvalues of J)

- converges to a limiting distribution $\lim_{N \rightarrow \infty} \rho_J(\lambda) = \langle \rho_J(\lambda) \rangle_J \equiv \rho(\lambda)$



 $g(\lambda)$
 $g_J(\lambda)$ for a single J at large N

- how do we average $g_J(\lambda)$ over J ? it depends on eigenvalues $\lambda_1, \dots, \lambda_N$ which are very complex functions of J

- work instead with Stieltjes transform:

$$\begin{aligned}
 \mathcal{E}_J(z) &\equiv \frac{1}{N} \text{Tr} [zI - J]^{-1} \\
 &= \frac{1}{N} \sum_{i=1}^N \frac{1}{z - \lambda_i}
 \end{aligned}$$

$\mathcal{E}_J(z)$ is Stieltjes transform of eigenvalue spectrum of J

here $z \in \mathbb{C}$

$\forall z \neq \lambda_i$ for any i this is an analytic function of z and it has simple poles at eigenvalues of J

- $\mathcal{E}_J(z)$ is a slightly simpler function of J than eigenvalues

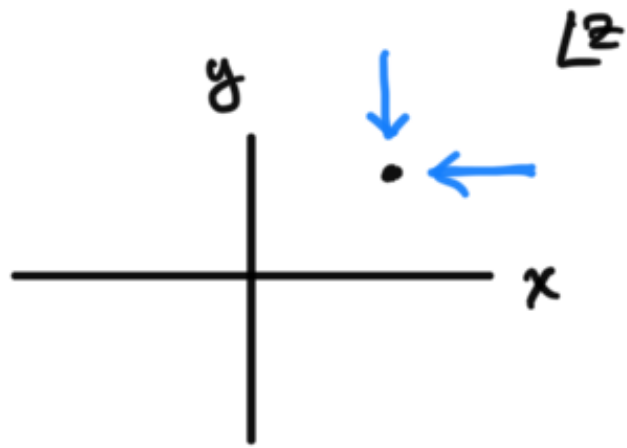
- if you know $\mathcal{E}_J(z)$ you can invert it to get $g(\lambda)$ via the inversion formula:

$$g_J(\lambda) = \frac{1}{\pi} \text{Im} \mathcal{E}_J(\lambda - i\varepsilon) \text{ as } \varepsilon \rightarrow 0$$

Intuition behind Stieltjes transform and eigenvalue distribution via electromagnetism in 2dim

how should we think about complex analytic functions $\varepsilon(z)$ in general?

- $z = x + iy$
- $\varepsilon(z) \equiv \varepsilon_x(x, y) - i \varepsilon_y(x, y) \quad \varepsilon_x, \varepsilon_y \in \mathbb{R}$
- what are the conditions on $\varepsilon_x(x, y)$ and $\varepsilon_y(x, y)$ so that $\varepsilon(z)$ is analytic (only a function of z) and complex derivatives exist



if a complex derivative of $\varepsilon(z)$ exists at a point z_0 , it should be the same no matter how you approach z_0 in the complex plane

approach horizontally:

$$\begin{aligned} \frac{\partial \varepsilon}{\partial z} \Big|_{z_0} &= \lim_{h \rightarrow 0} \frac{\varepsilon(z_0 + h) - \varepsilon(z_0)}{h} \\ &= \frac{\partial \varepsilon}{\partial x} \Big|_{z_0} = \frac{\partial \varepsilon_x}{\partial x} - i \frac{\partial \varepsilon_y}{\partial x} \end{aligned}$$

approach vertically:

$$\frac{\partial \varepsilon}{\partial z} \Big|_{z_0}$$

$$= \lim_{h \rightarrow 0}$$

$$\frac{\varepsilon(z_0 + ih) - \varepsilon(z_0)}{ih}$$

$$= \frac{1}{i} \frac{\partial \varepsilon}{\partial y} \Big|_{z_0} = \frac{1}{i} \left[\frac{\partial \varepsilon_x}{\partial y} - i \frac{\partial \varepsilon_y}{\partial y} \right]$$

$$= - \frac{\partial \varepsilon_y}{\partial y} - i \frac{\partial \varepsilon_x}{\partial y}$$

• for vertical approach and horizontal approach to yield same slope: must have

real part:

$$\frac{\partial \varepsilon_x}{\partial x} = - \frac{\partial \varepsilon_y}{\partial y}$$

$\Rightarrow$

$$\frac{\partial \varepsilon_x}{\partial x} + \frac{\partial \varepsilon_y}{\partial y} = 0$$

$\Rightarrow$

$$\vec{\nabla} \cdot \vec{\varepsilon} = 0$$

imag part:

$$- \frac{\partial \varepsilon_y}{\partial x} = - \frac{\partial \varepsilon_x}{\partial y}$$

$\Rightarrow$

$$\frac{\partial \varepsilon_x}{\partial y} - \frac{\partial \varepsilon_y}{\partial x} = 0$$

$\Rightarrow$

$$\vec{\nabla} \times \vec{\varepsilon} = 0$$

Cauchy Riemann Equations

conditions on $\varepsilon_x(x,y)$
and $\varepsilon_y(x,y)$

such that

$$\varepsilon(x,y) = \varepsilon_x(x,y) - i \varepsilon_y(x,y)$$

is an analytic function of

Electromagnetism in
a vacuum in 2 dimensions

• think of $(\varepsilon_x, \varepsilon_y)$
as a vector field in
 $\mathbb{R}^2$ instead of real
and imaginary parts of
a complex analytic
function

$$z = x + iy$$

ie $\varepsilon(x, y) \rightarrow \varepsilon(z)$

function $\varepsilon(z)$

- the vector field is divergence free and curl free
- like electric field in vacuum but in 2d

Summary:

$$\varepsilon(z) \equiv \varepsilon_x(x, y) - i \varepsilon_y(x, y)$$

$\varepsilon(z)$ analytic
in z
in some region R

$(\varepsilon_x, \varepsilon_y)$ as a vector field
in $\mathbb{R}^2$ is harmonic (div and curl = 0)
in some region R

- now consider the complex function which shows up in the Stieltjes transform

$$\varepsilon(z) = \frac{1}{z - \lambda}$$

- it is analytic everywhere except at $z = \lambda$ $\lambda \in \mathbb{R}$
- what electric field does it correspond to?

• need to compute its real and imaginary parts

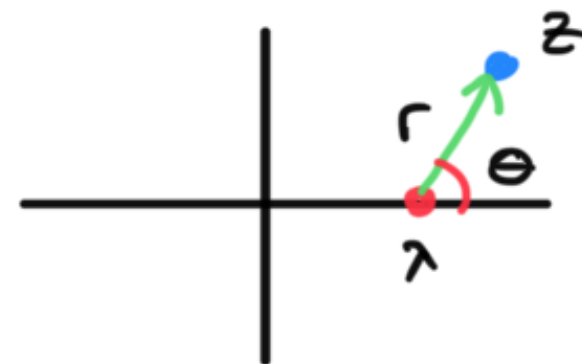
$$\varepsilon(z) = \frac{1}{z-\lambda} = \frac{1}{z-\lambda} \frac{z^*-\lambda}{z^*-\lambda}$$

$$= \frac{(z-\lambda)^*}{(z-\lambda)(z-\lambda)^*}$$

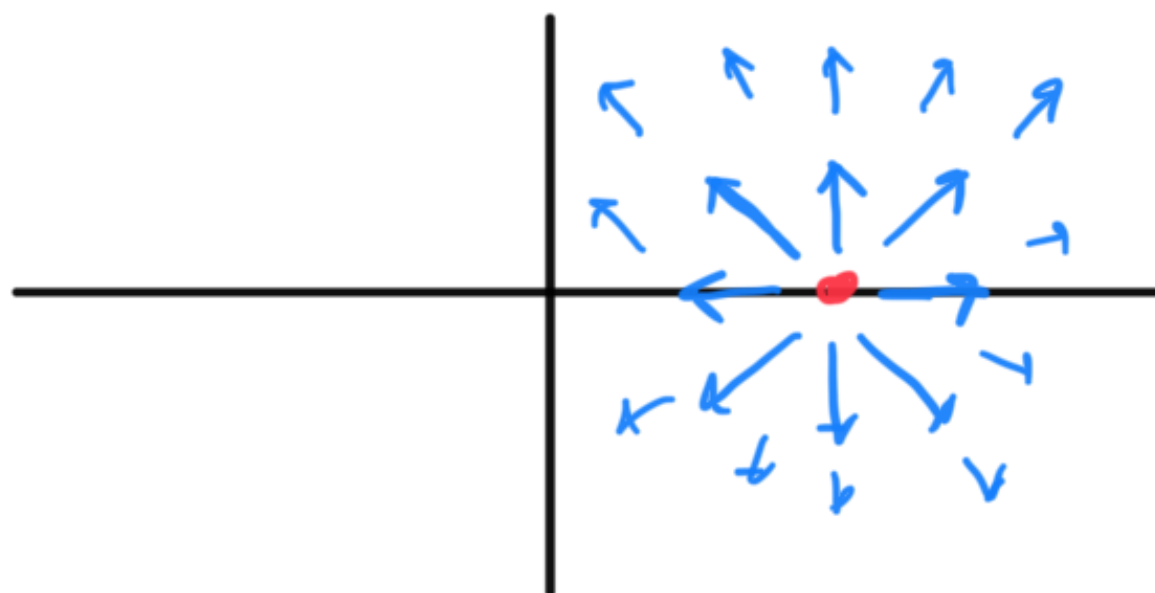
$$= \frac{r e^{-i\theta}}{r^2}$$

$$= \frac{\cos \theta - i \sin \theta}{r}$$

$$\text{let } z-\lambda = r e^{i\theta}$$



so $\varepsilon(z) = \varepsilon_x - i \varepsilon_y$ where $\varepsilon_x = \frac{\cos \theta}{r}$ $\varepsilon_y = \frac{\sin \theta}{r}$

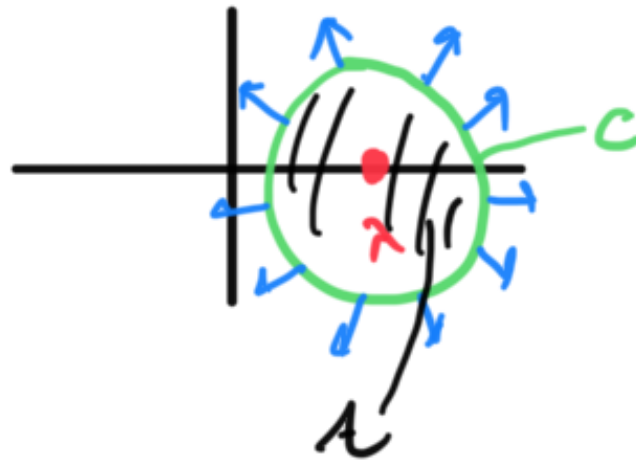


$$\varepsilon(z) = \frac{1}{z-\lambda}$$



electric field of a point charge
located at $z=\lambda$ in 2D complex plane

consider the flux of the electric field through a contour around it



$$\Phi = \int_C \vec{\varepsilon} \cdot d\mathbf{e}^\perp$$

$$\vec{\varepsilon} = \frac{\cos \theta}{r} \hat{x} + \frac{\sin \theta}{r} \hat{y}$$

$$d\mathbf{e}^\perp = \cos \theta \hat{x} + \sin \theta \hat{y}$$

$$\vec{\varepsilon} \cdot d\mathbf{e}^\perp = \frac{1}{r}$$

$$\Phi = \int_C \frac{1}{r} = (2\pi r) \left(\frac{1}{r} \right) = 2\pi$$

divergence theorem:

$$\Phi = \int_C \vec{\varepsilon} \cdot d\mathbf{e}^\perp = \int_A \vec{\nabla} \cdot \vec{\varepsilon}$$

$$\Rightarrow \vec{\nabla} \cdot \vec{\varepsilon} = 2\pi \delta(z-\lambda) \quad \text{"charge is } 2\pi \text{"}$$

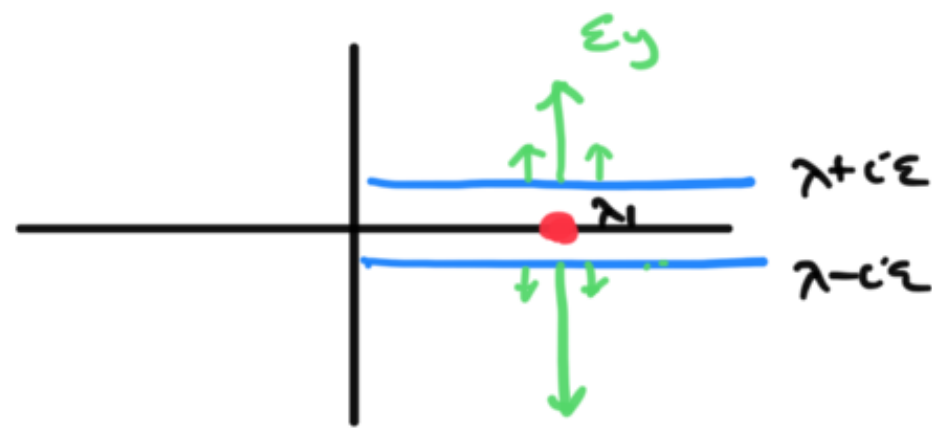
Now recall inverse formula:

$$\Sigma_J(z) = \frac{1}{N} \sum_{i=1}^N \frac{1}{z - \lambda_i}$$

↔ electric field from a bunch of point charges (normalized by $1/N$)

$$g_J(\lambda) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \operatorname{Im} \Sigma_J(\lambda - i\varepsilon)$$

visualize this for a single charge: $\Sigma_J(z) = \frac{1}{z - \lambda_1}$



$$\begin{aligned} & \operatorname{Im} \Sigma_J(\lambda' - i\varepsilon) \\ &= \operatorname{Im} [\varepsilon_x - i \varepsilon_y] \\ &= -\varepsilon_y (\lambda' - i\varepsilon) \\ &= \varepsilon_y (\lambda' + i\varepsilon) \end{aligned}$$

$$\lambda' = \lambda - \lambda_1$$

$$\begin{aligned} &= \frac{\sin \theta}{r} & r^2 &= (\lambda' + i\varepsilon)(\lambda' - i\varepsilon) \\ & & &= \lambda'^2 - \varepsilon^2 \end{aligned}$$

$$\sin \theta = \frac{\varepsilon}{r}$$

$$= \frac{\varepsilon}{\lambda'^2 - \varepsilon^2}$$

$$= \frac{\varepsilon}{(\lambda - \lambda_1)^2 - \varepsilon^2}$$

new

$$\delta[\lambda - \lambda_1] = \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \frac{\varepsilon}{(\lambda - \lambda_1)^2 - \varepsilon^2}$$

known fermions
result...

converges to
a δ -function

normalization y-component of electric field at $\lambda + i\varepsilon$
just above a point charge located
at λ_1

$$g(\lambda; \lambda_1, \varepsilon)$$

$$\lim_{\varepsilon \rightarrow 0} \int f(\lambda) g(\lambda; \lambda_1, \varepsilon) = f(\lambda_1)$$

for any differentiable $f(\lambda)$

Back to replica method

$$\rho_J(\lambda) = \frac{1}{N} \sum_{i=1}^N \delta[\lambda - \lambda_i]$$

• empirical eigenvalue density =
charge density of a set
of column charges in complex
plane

replica transform = electric

- $\Sigma_J(z) = \frac{1}{N} \sum_{i=1}^N \frac{1}{z - \lambda_i}$

$$= \frac{1}{N} \text{Tr} [zI - J]^{-1}$$

- $\Sigma_J(z)$ is the field of this charge density...

- we still don't know how to average the electric field $\Sigma_J(z)$ over J

- but what about the electric potential?

$$\Sigma_J(z) = \partial_z \sqrt{J}(z)$$

$$\frac{1}{z - \lambda} = \partial_z \ln(z - \lambda)$$

$$\Rightarrow \sqrt{J}(z) = \frac{1}{N} \text{Tr} \ln [zI - J]$$

$$\sqrt{J}(z) = \frac{1}{N} \ln \det [zI - J]$$

note for any sym matrix A

$$\text{Tr} \ln A = \ln \det A$$

$$\text{Tr} \ln A = \sum_i \ln \lambda_i$$

$$\det A = \prod_i \lambda_i$$

but how do we average $\sqrt{J}(z)$ over J ?

$$\sqrt{J}(z) = \frac{1}{N} \ln \det [zI - J]$$

$$= -\frac{2}{N} \ln \det [ZI - J]^{-\frac{1}{2}}$$

$$= -\frac{2}{N} \lim_{n \rightarrow 0} \frac{\left\{ \det [ZI - J]^{-\frac{1}{2}} \right\}^n - 1}{n}$$

$$\ln z = \lim_{n \rightarrow 0} \frac{z^n - 1}{n}$$

$$= \lim_{n \rightarrow 0} \frac{\partial}{\partial n} z^n$$

$$= F \quad \text{if} \quad z^n = e^{nF}$$

$$\sqrt{J}(z) = -\frac{2}{N} \lim_{n \rightarrow 0} \frac{\partial}{\partial n} \left\{ \det [ZI - J]^{-1/2} \right\}^n$$

Now how to average over J ? use gaussian integral rep of $[\det A]^{-1/2}$:

$$[\det A]^{-1/2} = \int \frac{d\vec{s}}{(2\pi)^{N/2}} e^{-\frac{\vec{s}^T A \vec{s}}{2}} \quad \vec{s} \in \mathbb{R}^N$$

Why? : recall normalization of a gaussian of covariance C

$$\int \frac{d\vec{s}}{\sqrt{\det C} (2\pi)^{N/2}} e^{-\frac{\vec{s}^T C^{-1} \vec{s}}{2}} = 1$$

$$\Rightarrow \int \frac{d\vec{s}}{(2\pi)^{N/2}} e^{-\frac{\vec{s}^T C^{-1} \vec{s}}{2}} = \sqrt{\det C}$$

$$\text{let } A = C^{-1}$$

$$\int \frac{d\vec{s}}{(2\pi)^{N/2}} e^{-\frac{\vec{s}^T A \vec{s}}{2}} = \sqrt{\det A^{-1}} = \frac{1}{\sqrt{\det A}}$$

Now we have a recipe to apply the replica method:

• let

$$Z_J(z) \equiv \int \frac{d\vec{s}}{(2\pi)^{N/2}} e^{-\frac{1}{2} \vec{s}^T [zI - J] \vec{s}}$$

auxiliary
partition function

• compute

$$Z(z, n) \equiv \langle Z_J(z)^n \rangle_J$$

replicated partition function

•

$$\psi(z) = -\frac{2}{N} \lim_{n \rightarrow 0} \frac{\partial}{\partial n} Z(z, n)$$

electric potential of
the charge density at location z
averaged over J

note if $Z(z, n) = e^{nF(z)}$ then $\psi(z) = -\frac{2}{N} F(z)$

•

$$\varepsilon(z) = \partial_z \psi(z)$$

going from electric potential to
electric field at location z

... constant



$$\zeta(\lambda) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \operatorname{Im} \zeta(\lambda - i\varepsilon)$$

going from electric field
near real axis $y \rightarrow 0$
to charge density on real
axis...

In the next lecture we will implement this program....